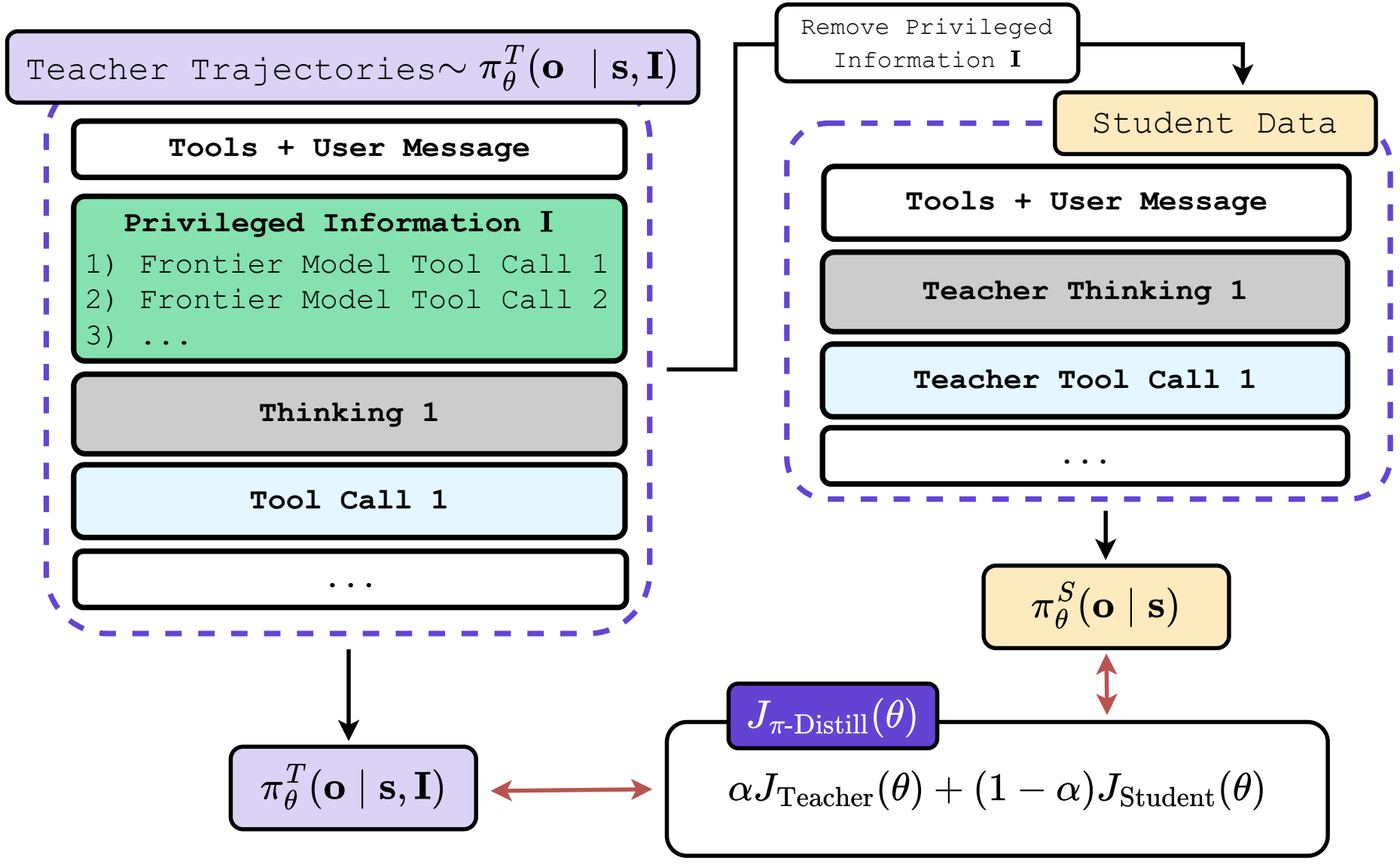
Teacher Trajectories $\sim \pi_\theta^T(\mathbf{o} \mid \mathbf{s}, \mathbf{I})$
Tools + User Message
Privileged Information I
1) Frontier Model Tool Call 1
2) Frontier Model Tool Call 2
3) ...
Thinking 1
Tool Call 1
...
Remove Privileged Information I
Student Data
Tools + User Message
Teacher Thinking 1
Teacher Tool Call 1
...
$\pi_\theta^S(\mathbf{o} \mid \mathbf{s})$
$\pi_\theta^T(\mathbf{o} \mid \mathbf{s}, \mathbf{I})$
$J_{\pi\text{-Distill}}(\theta)$
$\alpha J_{\text{Teacher}}(\theta) + (1-\alpha) J_{\text{Student}}(\theta)$